Victor Elkins

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Education

Georgia Institute of Technology
Master of Science, Computer Science (Human-Computer Interaction) | GPA: 3.90/4.00 — Graduated: December 2026

University of Houston
Bachelor of Science, Computer Science — Graduated: December 2024

Work Experience

IBM — Software Engineer — April 2025 – Present

- Led the evaluation and migration of IAM's production ingress from Application to Network Load Balancers, designing ALB vs. NLB load tests at up to 6,000 req/s (3× production peak) with 24-hour soak runs and custom Sysdig dashboards; traced recurring production timeouts to Akamai-driven connection spikes on the ALB.
- Led the NLB rollout across 4 staging and 22 production clusters in 7 regions for four IAM service teams, using fault-domain-phased deployments, cross-region traffic remapping, Envoy filter protections, and scripted CIS DNS cutovers.
- Drove PgBouncer connection pooling for IAM's PostgreSQL adoption: built control-plane/data-plane Helm charts, Istio egress routing, Secrets Manager integration, and Tekton/Jenkins automation for image-digest sourcing, delivering an end-to-end proof of concept and a deployment path through dev, staging, and production.
- Led performance hardening for the IAM Istio gateway, building a Go load-testing tool on Docker/Kubernetes with Jenkins-automated runs and worker scaling, and shipping data-driven rate limiting and TCP connection pooling that reduced Critical Incident Events, winning an **Outstanding Technical Achievement Award**.
- Optimized resource usage via data-driven metrics, migrating from IBM Cloud Classic to Red Hat OpenShift (ROKS); identified node inefficiencies to achieve approximately $250,000 in annual infrastructure cost savings.
- Architected a Blue/Green deployment strategy on the Dreadnought CD toolchain, automating cut-overs with Bash and refactoring Helm charts for zero-downtime releases.
- Built Context-Based Restrictions (CBR) features in Go, including Dedicated Instance API validation, storage, and VPC event handling, plus new rule-priority properties and validation.
- Strengthened production operations by eliminating recurring 500 errors from private-endpoint requests, removing noisy alerts based on 28-day telemetry, and remediating container CVEs across clusters while on call.

ExxonMobil — Software Engineer Intern — May 2024 – August 2024

- Designed a Go/Python-based API generator on Azure/OpenShift Kubernetes to automate secure Snowflake API creation, reducing development time by 90%.
- Engineered automated deployment pipelines using Apache Airflow and Datadog, ensuring secure and reliable deployments within the Azure cloud environment.

Dycom Industries — Software Engineer Intern — June 2023 – August 2023

- Developed a Python decision tree linter integrated into the GitLab CI/CD pipeline, automating code quality checks and significantly reducing manual review time.

Vertical AIT — Software Engineer Intern — February 2023 – June 2023

- Designed and deployed an internal image modeling tool using .NET and Azure DevOps, streamlining industrial processes and reducing manual tracking time by 95%.

Project Experience

Opioid Abuse Risk Prediction Portal — August 2025 – December 2025

- Developed a full-stack patient portal (React, Node.js) that performs CRUD operations against a FHIR-compliant database and integrates a machine learning model to predict a patient's risk of opioid abuse.

Award-Winning CNN for Breast Cancer Detection — August 2024 – December 2024

- Developed a PyTorch CNN model for breast cancer detection, incorporating ResNet and AlexNet architectures, which achieved second place at the University of Houston's AI and Data Science Showcase.

Skills

Languages: Go, Python, Java, JavaScript/TypeScript, SQL, Bash
Cloud & DevOps: Kubernetes/OpenShift, Istio/Envoy, Docker, Helm, CI/CD (Jenkins, Tekton), NLB/ALB, Akamai, Azure
Data & Observability: PostgreSQL, PgBouncer, Redis, Snowflake, Sysdig, Datadog, Prometheus/PromQL
Frameworks & Tools: React, Node.js, PyTorch, Git